令和７年度　工学部　入学試験問題

英　　語

（前　期）

注意事項

1　問題冊子１冊，解答用紙１枚が配られているか確認すること。

2　試験開始の合図があるまで，この問題冊子を開いてはいけない。

3　試験中に問題冊子（18ページよりなる）の印刷不鮮明，ページの落丁・乱丁および解答用紙の汚れ等に気付いた場合は，手を挙げて監督者に知らせること。

4　解答は解答用紙の所定の欄に記入すること。

5　解答用紙上部には**2箇所受験番号**を記入する欄がある。試験開始後直ちに記入すること。

6　問題冊子の余白等は適宜利用してよいが，どのページも切り離してはいけない。

7　試験終了後，問題冊子は持ち帰ること。

空白

I　次の英文を読み、問いに答えなさい。

The world is facing a significant food waste problem, with up to half of all fruit and vegetables lost somewhere along the agricultural food chain. Around 14% of food produced is lost after harvesting before it reaches shops and supermarkets globally.

Alongside food prices (66%), food waste is a concern for 60% of people that participated in a recent survey published by the *UK Food Standards Agency. (1) Other research suggests that as much as 25% of apples, 20% of onions and 13% of potatoes grown in the UK are destroyed because they look strange. This means that producers' efforts to meet strict requirements from buyers can lead to perfectly *edible produce being [a] discarded before it even leaves the farm – simply because of how it looks.

When imperfect fruit and vegetables don't make it to supermarket shelves, it can be due to standards for outward appearance. Supermarkets and consumers often prefer produce of a fairly [b] standard size that's free of *blemishes and other imperfections. This means fruit and vegetables that are *misshapen, discolored, or even too small or too large, are rejected before they make it to supermarket shelves.

In recent years there has been a growing trend of selling such "ugly" fruit and vegetables, both by major supermarket chains, as well as some *retailers that sell boxes of strangely shaped produce. And research has shown that 87% of people say they would eat imperfect fruit and vegetables if they were available. But other research indicates consumers can still be critical and difficult to predict. One study showed consumers are likely to throw away an apple with a blemish, but would eat a bent *cucumber.

So how can producers and retailers increase the amount of non-standard fruit and vegetable demand that not only reaches our shelves, but also our plates? Our recent research suggests a separate channel for selling such produce would increase profits for growers, lower prices for consumers, and improve overall demand for produce.

For growers, a *dedicated channel — either [c] independent or set up by a supermarket — to supply imperfect fruit and vegetables creates a new line of business. For retailers, this provides an opportunity for further revenue over and above current sales of standard produce to shops. When selling both types of

products to a single retailer, the ugly items might be undervalued compared with the standard-looking products. Our research also shows that selling the ugly produce through a dedicated channel is likely to increase total demand for fruit and vegetables, while also decreasing on-farm loss.

On the other hand, the growing market for ugly fruit and vegetables could be an economic threat to traditional retailers. It encourages newcomers into the market and could also limit the availability of "regular" produce because growers could become less strict about ensuring produce to meet traditional standards.

Increasing demand for imperfect fruit and vegetables across the supply chain will require all participants to get involved – from grower to seller. Here are some steps the various [d] parties could take. First, education about the environmental and economic impact of food waste could happen through marketing campaigns, in-store displays and even social media. Second, supermarkets and other major food retailers could revise their standards for how their products should look to accept a wider range of produce, including imperfect fruit and vegetables. This would help reduce food waste by making sure more produce is able to be sold.

It is also a good idea that farmers and growers will sell non-standard produce directly to consumers through farmers' markets or subscription services. This allows consumers to purchase fresh, locally grown produce that might not meet regular standards for supermarkets but that is just as nutritionally [e] beneficial. Supermarkets and growers could also donate produce rejected for how it looks to food banks, shelters and other organizations that serve those in need. This would help reduce food waste while also providing healthy food to those who might not otherwise have access to it.

By implementing these solutions, the supply chain can reduce the amount of ugly or imperfect fruit and vegetables that are wasted, while also providing consumers with healthy, affordable produce, even in times of supply chain shortages.

Behzad Hezarkhani, et. al. "How 'ugly' fruit and vegetables could tackle food waste and solve supermarket supply shortages." *The Conversation*, March 7, 2023.

* UK Food Standards Agency 英国食品基準庁 *edible 食べられる *blemishes 傷、欠点 *misshapen 不格好な *retailers 小売業者 *cucumber キュウリ *dedicated channel 特化した（販売、流通）経路

問１　下線部（1）を日本語に直しなさい。

問２　文脈に照らして、下線をつけた[a]から[e]の語と最も近い意を表すものを、１から４の中から選び、番号で答えなさい。

[a] discard　1 remove　2 resign　3 resume　4 retain

[b] standard　1 perfect　2 regular　3 stable　4 moral

[c] independent　1 subordinate　2 public　3 self-governing　4 open-minded

[d] parties　1 groups　2 classes　3 types　4 shops

[e] beneficial　1 acceptable　2 favorable　3 considerable　4 reasonable

問３　次の１から７の英文のうち、本文の内容に照らして適切なものを４つ選び、番号で答えなさい。

1 More than half of the people in the UK care not only about food price but also about farm productivity.

2 It is a global issue that lots of farm products would be thrown away because of outer appearance.

3 Still now, all of the retailers that deal with farm products would not sell ones that look strange.

4 Consumers are likely to be critical about how fruit and vegetables look, but most of them are not totally against buying imperfect products.

5 Imperfect vegetables should only be sold through dedicated market channels, as their presence would never be favored by traditional retailers.

6 In order to sell imperfect farm products, it is necessary to raise people's awareness about the bad effects of food waste.

7 Utilizing the "ugly" farm products would diversify consumers' choice, sustaining our food supply.

空白

II　次の英文を読み、問いに答えなさい。

Clay Bolt first saw a *rusty-patched bumblebee in an insect collection at the Great Smoky Mountains National Park in 2014. Brown and yellow with a faded rust-colored patch on its stomach, the *fuzzy, small insect got him thinking about the species' uncertain status in the wild. Once common in flower-rich *grasslands across the eastern United States and southeastern Canada, this bee now buzzes about in only 0.1 percent of the area it once occupied—a trend documented among many other North American bee populations as well. There are nearly 4,000 native bee species in North America, including 47 species of bumblebees, and many are declining or are at risk of [1] extinction.

A natural history and conservation photographer, Bolt wanted to [2] highlight the bumblebees' difficult situation. He decided to start by focusing on the rusty-patched bumblebee, telling its story through photographs and a film. "It broke my heart to think that if something wasn't done to protect it, *specimens like that one would be all we had left," Bolt says. "I felt like I could use my photography and filmmaking skills to really bring [3] attention to this species declining in the darkness."

His efforts, and those of conservationists, paid off in this case. In 2017, thanks to an appeal placed by *the Xerces Society and *the Natural Resources Defense Council, the rusty-patched bumblebee became the only bee in the continental United States to get protection under *the Endangered Species Act.

Much attention has focused on the decreasing number of *honeybees, a nonnative species that's domesticated and heavily managed for their *pollination services in agricultural settings. But the loss of wild bees, which are also important crop pollinators and possibly better suited to help native plants reproduce, is less recognized.

Large-scale agricultural activities have contributed substantially to (ア) this decline. Changes of bees' natural *foraging and nesting habitat to corn and soybean fields, plus heavy *pesticide use, have not only decreased native bee populations but have also had a bad effect on the insects' reproduction and colony growth. In a laboratory experiment, for instance, spraying pesticides—one of the most widely used classes of *insecticides since their introduction in the 1990s—led to an 85 percent reduction in new buff-tailed bumblebee queen numbers. These chemicals are sometimes in water and can be easily taken up by plants from the soil. "Every

time a bee eats pollen, which is their only source of food, they're getting small amounts of the insecticides," says Rich Hatfield, a conservation biologist with the Xerces Society.

Scientists are also concerned about a *fungal pathogen, detected in several declining North American bumblebee populations, including the rusty-patched bumblebee. Commercial *breeding, which brings together tens of thousands of queen bumblebees from the wild to start colonies in close distances, may have allowed the naturally occurring pathogen to spread among and between species. "These colonies would get taken out to farms and the pathogen could spread to wild populations," says *entomologist Sydney Cameron at the University of Illinois Urbana-Champaign.

Extreme temperatures, driven by climate change, are also linked to bumblebee declines across North America. In a recent study, researchers found these bumblebees [4] missing from many locations where they had previously been recorded. Those locations experienced increased warming and more frequent unusually hot days over time.

On a cool September morning in 2015, Bolt noticed his first rusty-patched bumblebee in the wild, at University of Wisconsin at Madison's garden. Such pollinator-friendly and pesticide-free gardens with wildflowers are helping conserve native bees. The Xerces Society has region-specific pollinator plant lists that can help all gardeners, even at home, choose plants that are bee-friendly as well.

Bolt has since turned to photographing other native bumblebee species in danger. Both the Franklin's bumblebee, which was once found across northern California and southern Oregon and was last spotted in 2006, and the western bumblebee, whose numbers have declined since the 1990s, are on (イ) his list. "People tend to identify readily with large *fauna," he says. (ウ) "But over 99 percent of life on Earth is smaller than your finger, and those things are so incredibly important."

Adapted from Priyanka Runwal, "For World Bee Day, take a moment to appreciate native bees." *National Geographic*, May 20, 2021.

*rusty-patched bumblebee ラスティーパッチド・バンブルビー（北米大陸に固有の種であるマルハナバチの一種） *fuzzy けば立った *grasslands 草原 *specimens 標本 *the Xerces Society 米国に拠点を置く非営利の自然保護団体 *the Natural Resources Defense Council 自然資源防衛協議会 *the Endangered Species Act 絶滅危惧種法 *honeybees ミツバチ *pollination 受粉 *foraging 食料を探すこと *pesticide 農薬 *insecticides 殺虫剤 *fungal pathogen 菌類病原体 *breeding 繁殖 *entomologist 昆虫学者 *fauna 動物群

問 1 本文の内容に照らして、下線部 [1] から [4] の語の意味に最も近いものを選び番号で答えなさい。

[1]	extinction	1	failure	2	pollution	3	assessment	4	disappearance
[2]	highlight	1	stress	2	influence	3	improve	4	encounter
[3]	attention	1	command	2	awareness	3	attraction	4	experience
[4]	missing	1	absent	2	errorless	3	unexpected	4	real

問 2 本文の内容に照らして、下線部（ア）の原因とはならないものを 1 つ選んで番号で答えなさい。

1 change of bees' habitat to corn and soybean fields
2 heavy use of pesticide
3 harmful chemicals in the soil
4 conservation programs for wild plants

問 3 本文の内容に照らして、下線部（イ）の内容を表わすものとして最も適切なものを 1 つ選んで番号で答えなさい。

1 pollinator-friendly gardens assisting native bees
2 region-specific pollinator plants that can encourage gardeners
3 other kinds of wild bumblebee species at risk
4 Franklin's bumblebees that currently live all over northern California

問４ 下線部（ウ）を日本語に直しなさい。

問５ 本文の内容に照らして、適切なものを２つ選び番号で答えなさい。

1 The number of rusty-patched bumblebees has decreased and now accounts for only 0.1 percent of all American bees.

2 The rusty-patched bumblebees were protected by the Endangered Species Act in 2017, just like other American native bees.

3 The decline in wild bee populations is not as well-known as the decline in honeybee populations which are brought in for agriculture.

4 Commercial breeding can play a part in stopping the decline of bumblebees by starting colonies in close distances.

5 Scientists are worried that commercial breeding may cause the pathogen detected in bumblebees to spread among species.

6 Bumblebees moved to North America from the south in order to avoid extreme temperatures, but it didn't work well.

III 次の英文を読み、問いに答えなさい。

In 1803, Richard Trevithick, a British mining engineer, planned and built a steam locomotive. His engine actually pulled a short train of cars uphill on a coal-mine railway in Wales. In the years after Trevithick's locomotive, several others were built for use on various British coal-mine railways. The world's first common carrier railroad (that is, one for public use) to use steam power was the Stockton and Darlington in England. It was designed and built by George Stephenson and opened for public service in 1825. In 1829, in the famous locomotive trials held at Rainhill, England, on the Liverpool and Manchester Railway, (ア) the Rocket, built by Stephenson and his son Robert, showed its superiority as a steam locomotive. Though not the first such locomotive, it was the beginning of effective steam power.

American colonists also built short *tramways for carrying stone or coal. In 1804, Oliver Evans declared that he could make a steam carriage that would run 15 miles an hour on good, [1] level railways. As early as 1812, John Stevens began to speak for a new kind of "rail-way." He wanted one that would [2] furnish long-range transportation, linking distant sections of the nation. In 1815, Stevens obtained the first permit to build a railroad across New Jersey, but he was unable to raise the money needed to build it.

The first common carrier railroad to be built in the United States was the Baltimore and Ohio. It was authorized in 1827 and construction started on July 4, 1828. The first steam locomotive to run in the United States, the English-built Stourbridge Lion, made a trial trip over the tracks of the Delaware and Hudson Canal Company in Pennsylvania in 1829. The engine was too heavy for the track and the trip was not repeated. In the summer of 1830, service began on the Baltimore and Ohio, though the trains were mainly pulled by horses at its early stage. Finally, in December 1830, an American-built steam-powered locomotive, the Best Friend of Charleston, carried a train of cars on the tracks of the South Carolina Railroad. The new operation combined the four essentials of track, trains, mechanical power, and common carrier service. The railroad had come to America.

Started in England, which had large financial resources and a dense population, the railroad had been well developed and became useful even in short distances. However, in America, they faced different problems with the railroad. America in the early 1800s was a nation of great distances, widespread population, and limited

capital. (イ) Americans had to learn to build railroads for their own country by actual experience; they could not copy English methods.

During the 1850s, north-south routes were developed both east and west of the *Alleghenies. It was not until after the *American Civil War, however, that a permanent railroad bridge (as distinguished from a temporary wartime structure) was constructed across the Ohio River. After the Civil War, the pace of railroad building increased. The two Pacific railroads—one, the Union Pacific, building *westward from Omaha, Nebraska; the other, the Central Pacific, building *eastward from Sacramento, California—had been started during the war to help promote national unity. They were joined at Promontory, Utah, on May 10, 1869, completing the first rail connection across the continent. By 1870, when the railroad movement in the United States was 40 years old, there were 53,000 miles (85,000 kilometers) of railway line. By 1910, the network was largely complete and there was little further extension. In 1916, total railroad-line *mileage in the United States reached its highest point at 254,000 miles (409,000 kilometers).

After 1920, with the rapid expansion of *paved roads, much traffic was taken from the railroads by automobiles, buses, and trucks. As a result, the railroad network began to shrink as lines that could no longer [3] pay their way were abandoned. (ウ) By the end of the 1980s, railroad-line miles in the United States had dropped to about 150,000 miles. Some of the lines had been built to serve mines, forests, or other natural resources and were abandoned when the resources were [4] exhausted. Other lines had been built to serve an anticipated need that was never realized. Still other lines disappeared because the industries they had been built to serve entered a period of decline or relocated to other parts of the country.

Adapted from "Railroad," *Britannica Kids*, 1 Mar. 2024.

*tramways 路面鉄道 *Alleghenies アレゲーニー山脈(アメリカとカナダ東部に連なるアパラチア山脈の一部) *American Civil War 南北戦争(1861-1865 年) *westward 西へ *eastward 東へ *mileage 走行距離 *paved 舗装された

問１　本文の内容に照らして、下線部（ア）の説明として最も適切なものを選び、番号で答えなさい。

1　In the trials, it marked the beginning of steam locomotives in the United States.

2　It proved itself as a better steam locomotive compared to previous trains.

3　The first locomotive using steam power ran on the Liverpool and Manchester Railway.

4　It was named by Stevenson and his son after their interest in cosmology.

問２　本文の内容に照らして、下線部 [1] から [4] の語句の意味に最も近いものを選び、番号で答えなさい。

[1]　level　1　vertical　2　even　3　height　4　stage

[2]　furnish　1　provide　2　confine　3　embellish　4　split

[3]　pay their way　1　waste their money　2　spend much fare　3　support themselves　4　be in debt

[4]　exhausted　1　put down　2　went down　3　tied up　4　used up

問３　本文の内容に照らして、下線部（イ）の理由として適切<u>でない</u>ものを１つ選び、番号で答えなさい。

1　The geographical conditions in America and England were quite different.

2　In America, unlike in England, people tended to be dispersed over large areas of land.

3　America had less capital to build railroads than England.

4　American people needed to develop their own electric engine technology.

問４　本文の内容に照らして、下線部（ウ）の原因として最も適切なもの１つ選び、番号で答えなさい。

1　rapidly expanded paved roads

2　high prices of automobiles

3　bankruptcy of many railroad companies

4　change of taste for travelling

問５　本文の内容に照らして、適切なものを１つ選び、番号で答えなさい。

1　John Stevens' plan was approved for financing by the U.S. government, but he was not able to use the money effectively.

2　One of the original railroads in the U.S. ended up with only test runs because of the heavy engine parts, and it later resumed the service using horsepower.

3　The Union Pacific Railroad was extended from Nebraska to California, completing the company's first long distance railroad in the early 19th century.

4　Some of the railroad-lines were assumed to be necessary but were not made due to a lack of space for actual construction.

IV　次の会話文を読み、問いに答えなさい。

Interviewer : Hi. Thanks for coming to the interview today. It's nice to meet you.

Candidate: Yeah, well, thank you.

Interviewer: To begin with,　(　A　)

Candidate: Okay, well, I've always been interested in teaching in a language program like this, and, I graduated with a degree in English and Psychology eight years ago.

Interviewer: Oh, really, both?

Candidate: Yes, and then I landed my first job overseas in Japan.

Interviewer: That's pretty impressive. What did you do there? What kind of work?

Candidate: Well, I worked full time for a private language school in Tokyo for the first two years, and then I found a job at a community college.

Interviewer: Oh really?　(　B　)

Candidate: Well, I taught English mainly, and sometimes helped students in their cooking classes.

Interviewer: You taught cooking classes?

Candidate: Well, I didn't actually cook, but many of my students wanted to become chefs in restaurants, or they wanted to start their own restaurants, and in our area, there were many tourists, including those who came from foreign countries.

Interviewer: So, your students needed to learn English for tourists, didn't they?

Candidate: Exactly.

Interviewer: Wow. That's, interesting. Anyhow, why did you return to the United States?　(　C　)

Candidate: I've been back for, uh, probably about a year now.

Interviewer: Okay, so what brought you back then?

Candidate: Well, my father, after retiring from the company he had worked at for more than 20 years, established a consulting firm specializing in multinational companies.

Interviewer: So you joined it.

Candidate: Yes I did. It was when my contract with the community college was over.

Interviewer: So you decided to come back to US. Then, what did you do there?

Candidate: I was a secretary. Scheduling, examining documents, Web page updates.

Interviewer: Was it interesting?

Candidate: Uh, (D) While working in my father's firm, I also took part in a treatment center as a therapist for teenagers struggling with depression and other mental health issues.

Interviewer: Oh, that is fascinating.

Candidate: To be honest, I found myself bored while working as a secretary. I realized that I wanted to work with young people, returning to teaching.

Interviewer: Um, you know, to be frank, we have fifteen people who are applying for this position. You have a wide range of experiences, and our language school has accepted lots of students from Asia. But why don't you tell me three reasons (E).

Candidate: Um, well, first of all, uh, I understand that different students have different learning styles, and for that reason, I've used iPads, video, music, drama, role plays, and games while teaching.

Interviewer: That's good.

Candidate: And second, I have a background in academic and psychological counseling and advising skills that are often needed, you know, in working with international students.

Interviewer: Uh, yeah, that could be really helpful.

Candidate: So you know, many of them struggle with emotional turmoil, homesickness and so on.

Interviewer: Yeah, coming to a new place . . . So, what would you say would be a third reason?

Candidate: Oh yes, finally, I speak four different languages.

Interviewer: Four? Wow! What languages do you speak, apart from English? Haha.

Candidate: Well, I speak Spanish, Portuguese, and Japanese at an advanced level.

Interviewer: At an advanced level?

Candidate: Right, and I also speak Arabic at a high-intermediate level.

Interviewer: Well, that's very, very impressive. You do have an impressive resume.

Candidate: Thank you.

Interviewer: We will contact you on Friday and let you know whether or not you'll be coming back for the second round of interviews.

Candidate: Okay.

Interviewer: It's been really nice meeting you, and thank you for your time.

(A) から (E) の (　　) に入れる最も適切なものを１つ選び、番号を書きなさい。

(A) 1 would you tell us the most memorable episode you have had?
2 why don't you tell us a little bit about yourself?
3 we all gather here to learn about the language program.

(B) 1 So, exactly what did you do in your work there then?
2 What was your first impression about your work?
3 Did you enjoy learning while working at these schools?

(C) 1 It's not too late for coming back. 2 You should have been there.
3 How long have you been back?

(D) 1 that's the point. 2 I suppose it was wrong.
3 it's up to you.

(E) 1 why you could enjoy any other job.
2 why you would be the best person for this job.
3 why your expertise can be suitable for any job.

(F) から (J) の問いに最も適切な答えを１から３の中から選び、番号を書きなさい。

(F) What kind of job is the candidate applying for?
1 a language teacher 2 a therapist 3 a secretary

(G) What was the candidate's first job in Japan?
1 a language teacher at a community college
2 a language teacher at a marketing firm
3 a language teacher at a private school

(H) What would the candidate use while teaching a language?
1 videos 2 pantomime 3 paintings

(I) How many languages can the candidate use at an advanced level or higher?
1 three languages 2 four languages 3 five languages

(J) What is scheduled after the interview?
1 The candidate will be informed of the result of the interview.
2 The candidate will be hired by the company after the second interview.
3 The candidate will have to submit a revised resume for the next interview.

空白

V　(1)　から (10)　の　(　　)　に入れるのに最も適切な語を、1から4の中から1つ選び番号で答えなさい。

(1) (　　) where we are to where we want to be is a considerable distance.
1 Beyond　2 From　3 As　4 On

(2) Kathy worked as a lawyer, and remained (　　) and professional in her job, which finally led her to be one of the executives in her firm.
1 dull　2 disgusting　3 diligent　4 desultory

(3) Due to the heavy snow, the road to the village (　　) to be so dangerous that many people living there made an emergency call to the police.
1 realized　2 entailed　3 constructed　4 proved

(4) It seemed that (　　) good had happened to him recently, as he was in high spirits.
1 something　2 anything　3 nothing　4 everything

(5) Severe storms (　　) millions of people in the East areas in US over the weekend.
1 landed　2 threatened　3 travelled　4 acclaimed

(6) Police said the thieves were obviously well (　　) with the alarm system at the department store.
1 calculated　2 surveyed　3 acquainted　4 learned

(7) The plan for a new office building went (　　) regardless of local opposition.
1 back　2 ahead　3 behind　4 down

(8) Plan your schedule carefully so that you waste as (　　) time as possible.
1 much　2 short　3 few　4 little

(9) The tea kettle was ( ) whistling when they entered, and she rushed across the kitchen to grab it.

1 eventually 2 hardly 3 already 4 largely

(10) More than a hundred firefighters are still trying to ( ) the fire at the plant.

1 contain 2 sustain 3 maintain 4 detain

空白

空白